

PORTFOLIO PROJECT

Smart Device Wellness Analysis

How can a wellness technology company play it smart?

30

Fitbit Users

R

Language

FitBit Data

Source

Three Business Questions

01

Smart Device Trends

What patterns emerge from how people use fitness trackers day-to-day — activity, sleep, calories?

02

Customer Application

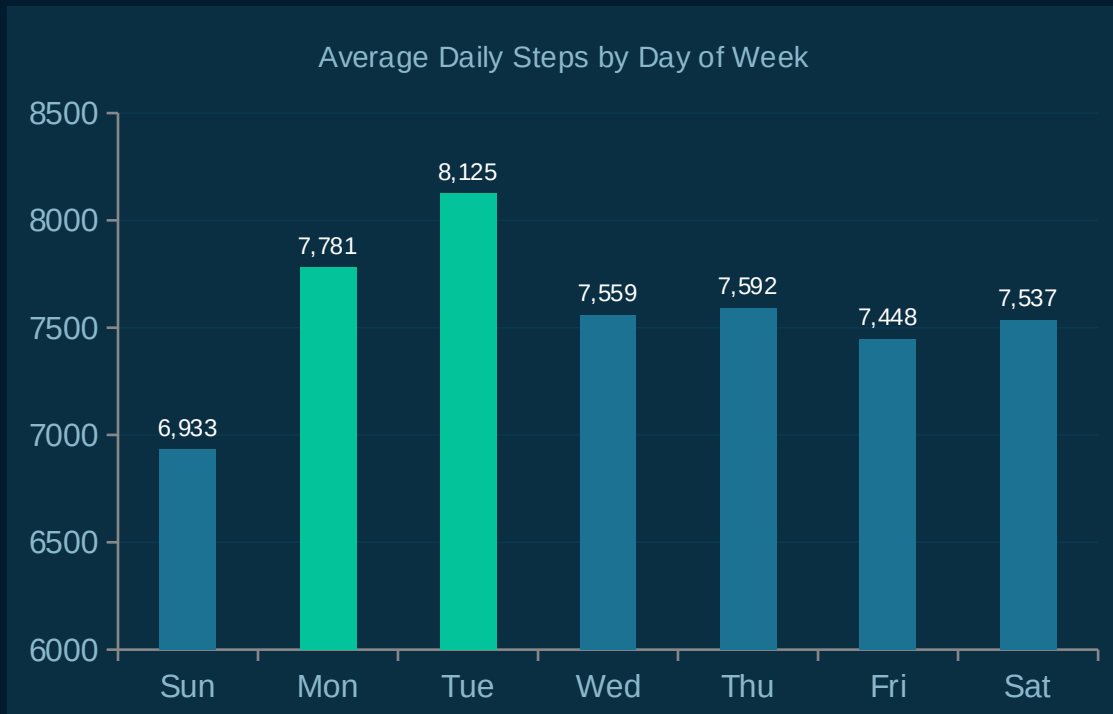
How do these smart device trends map to the behavior of Bellabeat's own wellness-focused customer base?

03

Marketing Strategy

How can these insights directly inform Bellabeat's marketing decisions to increase engagement and sales?

When Users Are Most Active



5-7 PM

Peak activity window after work hours

12 PM

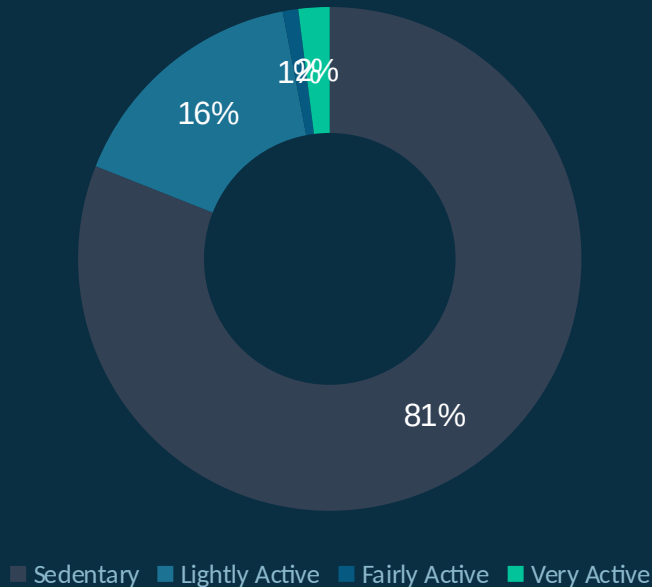
Secondary peak during lunch breaks

Sunday

Lowest step count — clear rest day pattern

Two Key Wellness Gaps

Average Day: Activity Breakdown



81%

of average day is sedentary — even among fitness tracker users

6.9 hrs

Average sleep — below recommended 7–9 hours for adults

Strong link

More time in bed = more actual sleep logged (near-perfect correlation)

3 Data-Driven Actions for Bellabeat

01

Target Peak Hours with Push Notifications

Users are most active 5–7 PM and 12 PM. Schedule motivational reminders and feature promotions during these windows to maximize engagement.

02

Build a Sleep Improvement Campaign

Average sleep is below recommended levels. A Bellabeat campaign focused on 'better sleep starts with better habits' — linking the Leaf and Time devices to bedtime reminders — directly addresses a proven gap.

03

Gamify Movement During Sedentary Hours

81% of the average day is sedentary. Introduce hourly movement nudges and weekly streak challenges in the Bellabeat app to break sitting patterns and increase daily engagement.

TOOLS USED

- ▶ R (tidyverse, ggplot2, dplyr)
- ▶ Visual Studio Code
- ▶ FitBit Fitness Tracker Dataset (Kaggle / Möbius)
- ▶ ROCCC framework for data quality assessment

Have a dataset and a business question? Let's talk.

One-time delivery · No maintenance required · Results in plain English